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REVIEW

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A review of deep learning transformer variants used in solar PV forecasting

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Abstract

Solar power has become one of the most important renewable energy sources as it is sustainable, its prices are reducing, and it could contribute to the transition to clean energy across the world. Nevertheless, the inherently intermittent nature of solar PV output due to the rapidly fluctuating weather conditions demands a very serious need to have accurate forecasting systems. The current reviews are primarily over traditional deep learning or statistical models, creating a knowledge gap on how transformer-based architectures intend to separate long-term dependencies and nonlinear behaviours of solar PV forecasting. Consequently, the objective of this review is to fill this gap by giving a detailed analysis of the variants of transformers that would be utilized when predicting PV power. A large variety of transformer models, among which are Simple Transformer, Informer, Temporal Fusion Transformer (TFT), Patch-Transformers, TSformer, and Hybrid Transformer models, among other recent models, are systematically analysed in this paper. Their architectural innovations, approach enhancement, benefits, and drawbacks in the forecasting of solar energy are described in the review. The strategy adopted also involves comparative appraisal, architecture-based classification, and analysis based on published benchmark investigations in terms of performance metrics. The result of this review demonstrates that transformer-based models are better than classic AI algorithms to capture long-range temporal patterns, process heterogeneous inputs, and achieve better forecasting results. The findings of the literature show further that strong increases in robustness, interpretability, and predictability of multiple horizons are achieved. Altogether, this review provides a unified picture of the modern transformer-based developments and defines the future research that ought to facilitate the improvement of scalability, uncertainty quantification, and the deployment of PV forecasting systems in the real world.

Keywords Deep machine learning, Solar PV forecasting, Temporal fusion, Transformer, Informer transformer, Hybrid transformer model, Patch- transformer

Introduction

Solar photovoltaic (PV) power generation has become one of the most important elements in the clean energy conversion worldwide because of its sustainability, scalability, and the rapidly dropping prices of installation. The nature of the intermittency provided by weather conditions, clouds, shifts in temperatures, and changes in seasons, however, poses serious problems in maintaining consistent and predictable power supplies. Solar PV predictability is thus critical to grid plans, energy management, load balance, and permits smarter energy management mechanisms.

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Common prediction methods, such as statistical models, early deep learning networks (RNN, LSTM, GRU, CNN), do not always have access to long-range non-linear temporal correlations of PV generation data. These constraints have seen growing attention to transformer-based architectures, which have already shown impressive success in sequential modeling with self-attention mechanisms and parallel processing abilities.

In 2017, Vaswani et al. created the Transformer model that transformed the essence of natural language processing (NLP) with its original process of self-attention and the possibility to process information concurrently. Unlike recurrent neural networks (RNNs) that process sequential data one after another (step-by-step), the Transformer processes long continuous sequences simultaneously to allow it to capture long-range associations more effectively and efficiently (Vaswani 2017). The Transformer model has the following key components: the self-attention mechanism, multi-head attention, encoder and decoder layers, and positional encoding. The self-attention process is also used so that the model can determine the importance of different elements in the sequence of inputs in processing each element (Yang et al., 2024a). This is realized through the calculation of the attention weight according to the correlation between two groups of elements of the series. Attention weights are subsequently used on the input elements to produce a weighted sum of the elements that constitute the context-sensitive representation of each element (Zhu et al., 2023b).

Multi-head attention is based on a self-attention mechanism that allows the model to be trained on multiple groups of attention weights that characterize several attributes of the correlation between the elements in the sequence (Phan et al., 2024). This strengthens the ability of this model in capturing complex dependencies, which eventually improves the performance of the model (Dong et al., 2024). Transformer architecture consists of an encoder and decoder comprising various layers (Yang et al., 2024a). The input sequence is processed by an encoder to obtain a contextualized representation, and the output sequence is produced by a decoder using a representation. Self-attention and feed-forward neural networks, residual connections, and layer normalization are provided on every encoder and decoder level to facilitate the training process, as well as to improve the performance (Phan et al., 2024).

The Transformer model is fed with positional encoding information (Yang et al., 2024a) on the position of the respective elements of the input sequence. Seeing the self-attention mechanism, the permutation-invariant mechanism, positional encoding is needed to encode the time of the data sequence. Several positional encoding encodings have been proposed, including learned embeddings and sinusoidal encoding functions (Vaswani 2017). The cumulative dependencies that the Transformer model represents, its parallel execution capability, and its modular design have led to its widespread adoption in numerous time-series forecasting applications, including solar PV power prediction (Vaswani 2017). Furthermore, the Tensor2Tensor framework introduced by Vaswani et al. (Vaswani 2018) facilitated the implementation and extension of Transformer-based architectures, accelerating research and development of advanced attention-based models across a wide range of forecasting domains. As a result, several transformer variants have been developed to address specific challenges in time-series forecasting, including long sequence lengths, non-stationary dynamics, multivariate inputs, and model interpretability. These variants are discussed below.

Although transformer-based architectures have gained considerable attention in recent years for time-series forecasting, the existing literature still lacks a consolidated and detailed review focused specifically on their application in solar PV power prediction. The available reviews still focus primarily on conventional deep-learning-based systems (LSTM, GRU, and CNN) or are quick descriptions of transformers with no focus on their architecture both limitations, and domain-specific issues. Very minimal literature exists that provides a comparison of all prominent variants of transformers, regardless of the newer models, such as Patch-Transformer, TSformer, CT-PatchTST, WECT, and SL-Transformer, in solar PV forecasting (Hertel et al., 2022) (Yang et al., 2024a) (Zhu et al., 2023a). This also poses a call to research void since there is no exhaustive study which has been conducted to explain how these models deal with issues like long-term dependence, non-stationary solar data, multiple channel weather inputs, interpretability, and strong multi-horizon prediction (Abdullah et al., 2020; Raina et al., 2023).

Since the architecture of transformers has grown at an alarming pace and is undergoing more and more diversity, a review of transformer architecture is urgently required to make researchers aware of the advantages, disadvantages, and differences in methodologies of each type of architecture. The effects that are unique to solar PV forecasting include the existence of highly dynamic weather patterns, multivariate inputs, and large-scale time-series behavior, which cannot be generalized well based on reviews on generic forecasting tasks (Gupta et al., 2021; Santos et al., 2022). Thus, there exists a high demand for a consistent and coherent overview of reviews on transformer models about solar PV forecasting, and in this regard, it permits researchers and practitioners to make accurate decisions regarding models to select, data set requirements, computation requirements,

and scope of application (Phan et al., 2024; Hertel et al., 2022).

While foundational reviews by Hendouzi & Bourouhou (2020) and Massaoudi et al. (2021) provided comprehensive overviews of the transition from statistical methods to first-generation deep learning models (e.g., RNNs, LSTMs, and basic CNNs), they primarily focus on general AI performance metrics.

Since that time, the field has experienced a revolution of transformers that have defined special-purpose architectures to overcome the fundamental shortcomings of generic self-attention. In this review, the authors make three different contributions:

Temporal and Generational Scope: Compared to prior research on the analysis of models up to 2021, this study considers studies on Second-Generation Transformer variants published between 2023 and 2025—including but not limited to Patch-Transformer, TSformer, and CT-PatchTST—that did not exist as of the earlier survey periods.

Mechanism-to-Challenge Mapping: This work gives detailed architectural consideration to how particular mechanisms (e.g., ProbSparse attention, Variable Selection Networks, and Series Decomposition) are explicitly designed to solve domain-specific PV problems such as non-stationarity, multivariate weather intermittency, and others.

Scenario-Based Decision Framework: In addition to a list of literature, this paper develops a new framework of comparative concept (Table 6) which projects transformer variants to particular scenarios of practice in the industry (e.g., noisy data, multi-site forecasts, or real-time demands) that can guide the technical choice of the practitioner (Hendouzi & Bourouhou, 2020; Islam et al., 2023; Mao et al., 2024).

The main objectives of this review paper are:

- To provide a review of transformer-based deep learning architectures used specifically for solar photovoltaic (PV) power forecasting (Hendouzi & Bourouhou, 2020; Islam et al., 2023).
- To analyse and compare architectural features, attention mechanisms, and learning strategies employed by major transformer variants.
- To evaluate the advantages, limitations, and performance differences among models such as Informer, TFT, Patch-Transformer, TSformer, CT-PatchTST, WECT, SL-Transformer, and hybrid transformer frameworks (Huo et al., 2025; Wang & Kang, 2024).
- To highlight how these architectures address solar PV forecasting challenges, including non-stationary, multi-variate weather inputs, long-term dependencies, and interpretability (Hertel et al., 2022; Yang et al., 2024a).
- To create a scenario-based model selection guide that assists researchers to select the most optimal variant of the transformer to use to complete short-term, long-term, seasonal, and noisy data forecasting tasks.
- To outline future research directions**,** including uncertainty quantification, real-time forecasting, hybrid model design, and integration of numerical weather prediction (NWP) data (Hu et al., 2024; Wang & Kang, 2024).

Transformer model (in solar PV forecasting), its structural components, its attention ability, its interpretability, and its ability to learn long sequences are well addressed in this paper. It wraps up the performance results of various datasets, forecast horizons, and evaluation measures and gives a clear picture of both the strengths and weaknesses of all the models. In addition, this review offers practical details on the ways transformers can use multivariate weather inputs, trend-seasonality structures, and even the temporal dependence with regard to PV forecasting. By conducting a comprehensive study of architecture, comparison and future research recommendations. This study turns out a comprehensive tool to both experts and industry stakeholders in the endeavor of using the advanced transformer models to predict solar energy (Gupta et al., 2021; Hendouzi & Bourouhou, 2020).

The process of solar PV power forecasting involves obtaining the meteorological data, advance generation of PV, and other environmental variables and then utilizing the best models, which are high-quality AI models, namely, transformer architectures. The outputs are long-term or short-term PV power predictions, which are vital in the effective operating of a grid and energy planning. Figure 1 presents the general work flow in transformer-based solar PV power forecasting indicating correlation of the data inputs and forecasting models, forecast output.

Systematic review methodology

To promote transparency, rigor and reproducibility, this review was carried out in strict compliance with PRISMA (Preferred Reporting Items to Systematic Reviews and Meta-Analyses) guidelines. This methodology was created to avoid bias and has offered a broader picture of the shift towards modern deep learning to enhanced Transformer-based architectures in the area of renewable energy prediction.

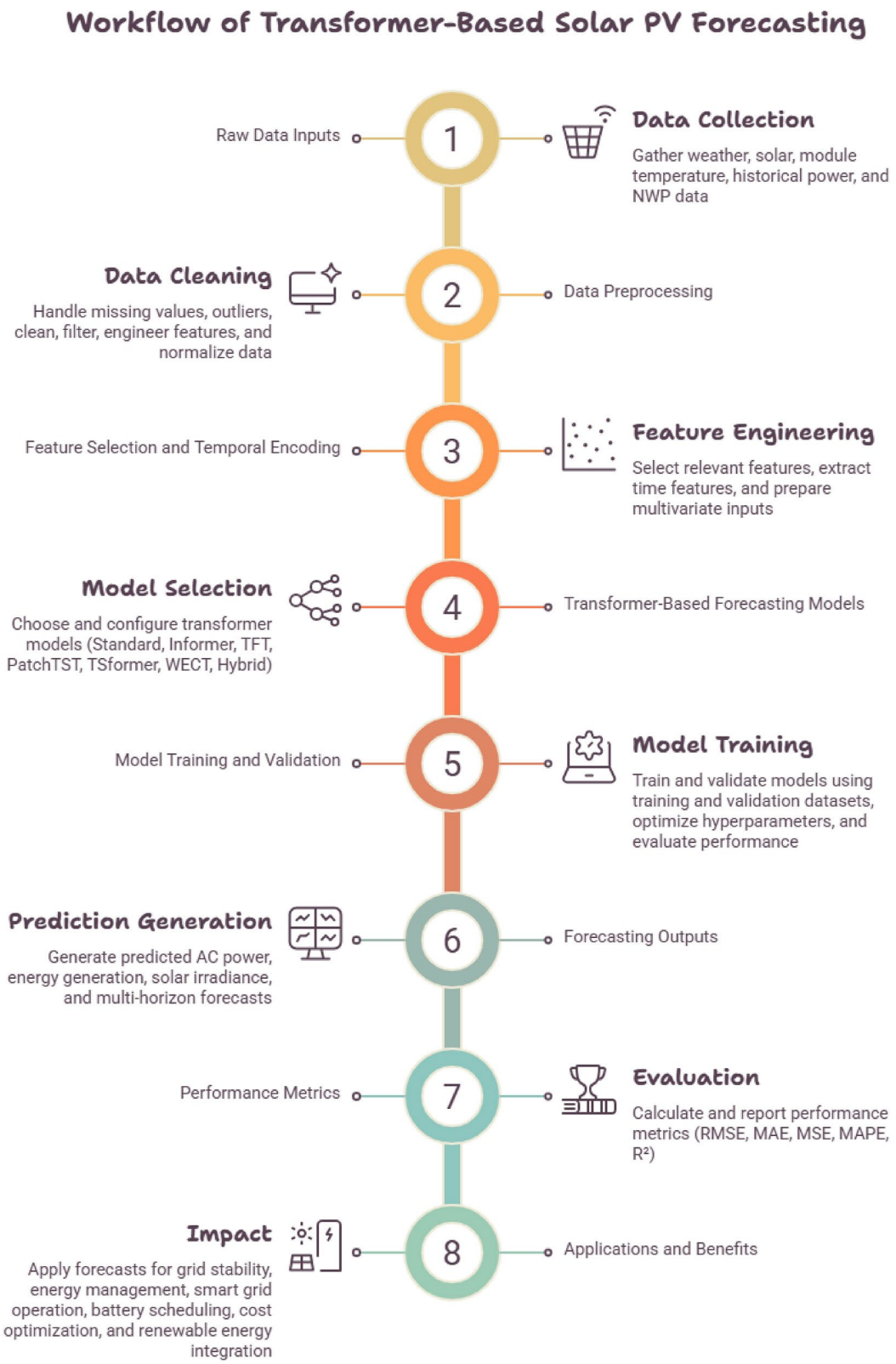


Fig. 1 Work flow diagram for transformer-based solar PV power forecasting. (Source: Created by the author using Napkin AI.)

Search strategy and databases

A systematic search was conducted in the significant academic databases, such as IEEE Xplore, ScienceDirect (Elsevier), MDPI, SpringerLink and Google Scholar. To encompass the evolution of classic architectures to the latest time-series standards in modern times, the search period was between 2017 (the launch of the Transformer model) and early 2025 (Zheng et al., 2024) (Wen, et al., 2023). We searched using the following main keywords and Boolean operators:

1. ("Transformer" OR "Attention Mechanism") AND ("Solar PV Forecasting" OR "Photovoltaic Power Prediction").
2. ("Temporal Fusion Transformer" OR "Informer" OR "TSformer") AND ("Time-series Forecasting").
3. ("Solar Irradiance" OR "Renewable Energy") AND ("Deep Learning Variants").

Inclusion and exclusion criteria

To keep the technical standards high and have the issues investigated in the review centered on the actionable, empirical data, eligibility criteria were developed. These criteria formed the guide of full-text screening stage and saw to it that only high quality and relevant research was synthesized. The following criteria were used:

- Inclusion Criteria: (1) Peer-reviewed journal articles and conference proceedings; (2) Studies explicitly implementing Transformer-based architectures for solar PV or irradiance; (3) Research providing empirical performance metrics (e.g., RMSE, MSE, MAE, R^2).
- Exclusion Criteria: (1) Studies focusing on general deep learning without a Transformer component; (2) Articles not written in English; (3) General NLP Transformer papers without time-series applications.

Article selection and screening

The process of the study selection was implemented following the PRISMA guidelines and visually presented in Fig. 2. A total yield of 450 records was obtained in the initial identification phase in which total searches were conducted in specific academic databases of interest. After excluding the duplicates and initial screening of titles and abstracts, 180 articles were selected to undergo a stringent and full-text eligibility test. Following architectural novelty, citation effects, and particularity in respect to the particular problems of solar energy forecasting,

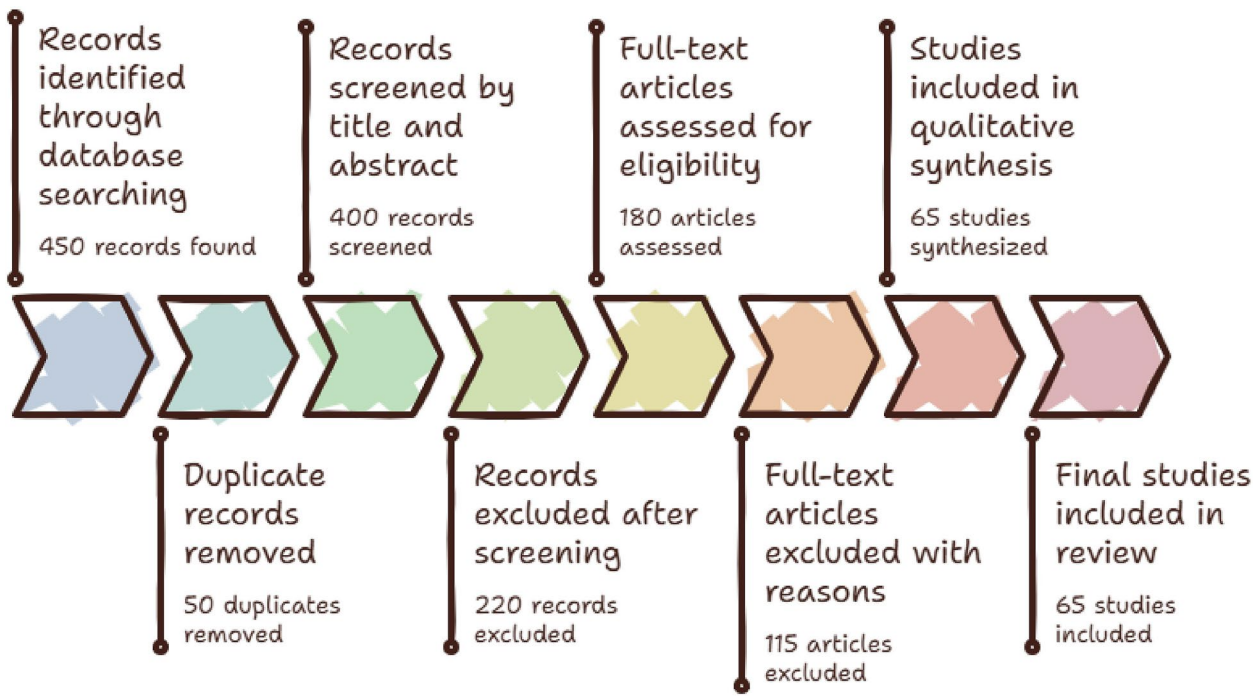


Fig. 2 PRISMA flow diagram and dual-investigation model that describes the method of systematic literature selection ($n = 65$) and integrated technical benchmarking methodology. (Source: Created by the author using Napkin AI.)

the eventual list of 65 main references, upon which this review was based, was subsequently obtained.

Foundational time-series transformer variants

A sequence of transformer variants bridges the step between general self-attention and specific solar forecasting, in the domain. The models were the first to leave the restrictions of the original NLP centric architecture by adding structural priors directly on time- series data. The next subsections describe the mathematical breakthroughs that set the existing technical setting of long-term time-series forecasting (LTSF).

Autoformer (decomposition and auto-correlation)

Autoformer transforms the Transformer into a deep decomposition. It sequentially removes trend cyclical and seasonal data using internal decomposition block. Importantly, it substitutes conventional self-attention with an Auto-Correlation scheme, which recognizes stochastic processes theory-based similarities of sub-series, and is much more efficient in long-term prediction (Wu et al., 2021; Kitaev et al., 2020; Katharopoulos et al., 2020).

In Eq. (1), Autoformer defines the process of full decomposition of the series X into trend-cyclical (X_t) and seasonal (X_s) in the following way:

$$X = SeriesDecomp(X) \rightarrow \{X_t, X_s\} \tag{1}$$

It replaces normal attention with Auto-Correlation from the Wiener-Khinchin theorem, which calculates the time-domain dependencies as defined in Eq. (2) (Li et al., 2019):

$$Auto - Correlation(Q, K, V) = \sum_{k=1}^{L} Roll(V, \tau_k) \cdot Softmax(Corr(Q, K)) \tag{2}$$

and it is with the top-k correlations that the time delays are represented by τ_k.

FEDformer (frequency domain analysis)

This model is mostly used in the frequency domain to form a global structure and long-range dependencies. It makes use of Frequency Enhanced Blocks (FEB) and Frequency Enhanced Attention (FEA) on the Fourier or Wavelet transforms. It bypasses the quadratic complexity of the conventional attention since it concentrates on a subset of the frequency spectrum but at high accuracy (Zhou et al., 2022).

FEDformer uses the frequency domain method of performing attention to capture overall structures in the order of the complexity of *O(L).* It follows a Frequency Enhanced Attention (FEA) is specified in Eq. (3):

$$FEA(Q, K, V) = \mathcal{F}^{-1}(Select(\mathcal{F}(Q) \odot \mathcal{F}(K)) \cdot \mathcal{F}(V)) \tag{3}$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ are the Fourier Transform and its inverse, allowing the model to focus on the most impactful frequency components (e.g., daily solar cycles).

iTransformer (architectural inversion)

iTransformer overcomes the drawback that Multivariate dependencies are difficult to deal with in standard Transformers. It works in the opposite direction of inverting the architecture by tokenizing variates (whole series) instead of each time step. This enables the explicit modeling of cross-variate correlations with self-attention, and Feed-Forward Networks (FFNs) respond to temporal variations, and thus a user-friendly basis of high-dimensional data (Liu et al., 2024).

iTransformer reverses the common methodology to tokenize the entire sequence of a single variate instead of a time step. Equation (4): Given a multivariate input $X \in R^{T \times C}$ (where T is time and C the number of variables, such as temp, irradiance, etc.) can be expressed in Eq. (4):

$$InvertedToken_i = Embedding(X_{:,i}) \in R^D \tag{4}$$

This can be used to explicitly model the inter-variate correlations between different sensors or weather parameters by the attention mechanism.

Crossformer (cross-dimension dependency)

Unlike all models, Crossformer allows cross-dimension dependency to exist (inter-variable dependencies). It employs a Dimension-Segment-Wise (DSW) embedding to generate a 2D vector array (time vs. dimension) and a Two-Stage Attention (TSA) layer to perform the dependency capturing between the two axes at the same time (Zhang & Yan, 2023).

TimesNet (2D variation modeling)

TimesNet models 1D time series as sets of 2D tensors using multi-periodicity (e.g., daily or annual solar cycles). This enables the model to represent intraperiod (within a cycle) and interperiod (between cycles) variations with typical 2D vision kernels (Inception blocks) making the analysis of complex temporal structures uniform (Wu et al., 2023).

TimesNet transforms 1D time series into 2D tensors based on multiple periods *P={P1……..Pk}.* For a period

Table 1 Comparison of foundational time-series transformer architectures

Model	Core innovation	Mathematical focus	Primary advantage
Autoformer	Auto-Correlation	Time-domain periodicity	Efficient long-term trend extraction
FEDformer	Fourier/Wavelet Blocks	Frequency-domain features	Global dependency capture with *O(L)* complexity
iTransformer	Axis Inversion	Cross-variate attention	Superior multivariate scaling & generalization
Crossformer	Two-Stage Attention	Cross-dimension correlation	Explicit modeling of inter-variable dependencies
TimesNet	1D to 2D Transformation	Multi-period variations	Unified capture of seasonal & cyclical patterns

p_i, the 1D series $X_{1:T}$ is reshaped into a 2D variation is define in Eq. (5):

$$X_{2D}^{(i)} = Reshape_{p_i, \frac{T}{p_i}}(X_{1:T}) \tag{5}$$

This enables the 2D kernels (art of Computer Vision) of this model to respond to complex intra-period and inter-period patterns in the sun.

These base variants (as summarized in Table 1) are a substantial breakthrough in computing efficiency and representational capability. Although these models have been first created as tools to perform generic forecasting activities, the fundamental innovations series decomposition, frequency analysis and inverted tokenization have become the architectural cornerstones of the solar-specific variants, examined in the following section. Knowledge of this lineage is important to understand the ability of existing PV-forecasting models to respond to the dual challenges of daily periodicity and high-frequency weather noise.

Different transformer variants

Standard transformer

A simple/standard transformer is a time series forecasting model built on the original Transformer architecture, one of which is used for solar PV power prediction (Orjuela-Can et al., 2024). It uses the self-attention mechanism to extract the temporal dependency of the PV power data so that the model can learn temporal patterns and relations that are paramount to forecasting accurately (Zhu et al., 2023b; Salman et al., 2024). Imposed on the overall input sequence simultaneously, the simple transformer can capture effectively long-range dependencies potentially missed by other time series models such as ARIMA or recurrent neural networks (RNNs), Vaswani et al. (Salman et al., 2024).

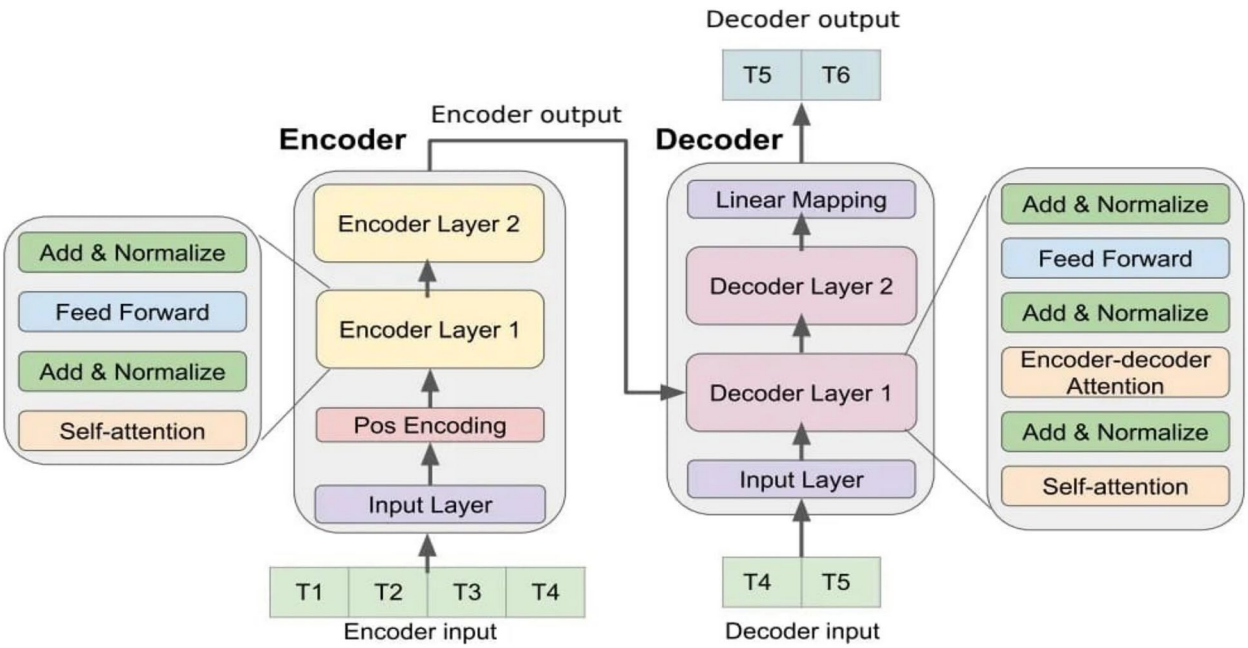


Fig. 3 Standard Transformer Architecture (Encoder–Decoder Model). (Source: Adapted from Salman et al. (2024; Salman et al., 2024))

Figure 3 (Vaswani, 2017), represents the basic Transformer architecture, which is an encoder-decoder architecture. The encoder acts on the input sequence tokens on the left side, and these are perceived to be T 1, T 2, T 3, and T 4. The encoding of word meanings and their position is achieved using the embedding layer, and the positional encoding is achieved by taking each token through that layer. The encoded tokens are processed using several encoder layers, first shown (two here). A self-attention mechanism, which allows every single token to notice all other tokens in a sequence, and an encoder feed-forward neural network are the two essential elements of each of the two encoder layers. All the connections between these two sub-layers are residual as well, and layer normalizing is used to stabilize learning. The final encoder is a contextualising embedding of the input sequence (Vaswani, 2017).

The decoder is the production of the target sequence, a sequence of bits, on the right. In the decoder, the already generated tokens (shifted to the right, e.g., [T 4, T5]) are received as input. The two (in the figure, there are shown two layers of the decoder are subdivided into three subsections. One, a masked self-attention mechanism gives predictions of a single position which are conditional on all the past tokens, but not the future ones. Second, encoder output information is added using an encoder-to-decoder attention circuit, and this enables the decoder to bias the predictions on the input sequence. Third, the feed-forward neural network then maps the features again by the residual connection's normalization (Vaswani, 2017).

Finally, the decoder output is passed through a linear mapping/softmax layer to produce word possibilities in the vocabulary, and outputs the following tokens (e.g., [T5, T6]) of the sequence. The cycle is repeated until the entire production sequence is produced.

The basic transformer, such as a multi-head attention as well as positional encoding to be more sensitive to temporally intricate patterns (Vaswani, 2017). Multi-head attention enables the model to estimate multiple sets of attention weights, including various aspects of the relations between time steps (Vaswani, 2017). Positional encoding also informs the model of the position of each time step within the sequence, and allows it to tell apart those time steps with the same value but a different temporal position Mao et al. (Mao et al., 2024).

The foundational Transformer architecture relies on the Scaled Dot-Product Attention mechanism. To define the attention mechanism, given an input sequence as queries *(Q),* keys *(K),* and values *(V)* with dimension d k, attention is defined mathematically in the Eq. (6):

$$Attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{6}$$

This enables the model-to-model long-range dependencies by computing the correlation between all-time steps of a sequence. Multi-Head Attention (MHA) is used to handle varying patterns in time, with the attention in this case operating in parallel over h distinct learned linear projections (defined in Eq. (7)):

$$\begin{aligned} &MultiHead(Q,K,V) \\ &= Concat(head_1,\ldots,head_h)W^O \end{aligned} \tag{7}$$

and where each of the attention heads is prescribed in Eq. (8):

$$head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{8}$$

This unified baseline provides the mechanism to capture the non-linear temporal correlation that is needed in data of PV generation.

Informer transformer

To address the limitation of the standard transformer as an adequate instrument in long-time series prediction, the Informer Transformer is developed with certain concerns to reduce the computation cost and facilitate efficacy (Walczewski & Whrle, 2024). The Informer brings two important models to the table, i.e., Prob Sparse attention and self-attention distilling operation.

Prob Sparse attention makes the computational burden of the self-attention mechanism linear instead of quadratic by attending only to the most important time steps (Song et al., 2024). This is done by applying a sparsity measure to determine the dominating queries, which are then applied to compute the weights in the attention. Prob Sparse attention can also search the long time series data of the Informer, which is inefficient, to a lesser extent, thus reducing attention calculations, which makes it significantly more efficient than the Informer (Walczewski & Whrle, 2024).

The standard attention in Transformer is defined in Eq. (9):

$$Attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{9}$$

However, in Informer, the ProbSparse attention focuses on dominant queries is defined in Eq. (10):

$$ProbSparseAttention(Q,K,V) = softmax\left(\frac{Q_{top}K^T}{\sqrt{d_k}}\right)V \quad (10)$$

where Q_{top} represents the top-u informative queries selected based on sparsity measurements.

The sparsity measurement is computed as Eq. (11):

$$M(q_i,K) = \max_j\left(\frac{q_i k_j^T}{\sqrt{d_k}}\right) - \frac{1}{L_K}\sum_{j=1}^{L_K}\frac{q_i k_j^T}{\sqrt{d_k}} \quad (11)$$

This equation identifies queries with the most significant attention distribution.

Self-attention distillation operation decreases the computational complexity even more, by gradually shortening the length of the sequence in the encoder (Zhu et al., 2023a). This is realized by down-sampling the attention maps, which automatically decreases the time steps that must be propagated in a later layer. The most valuable information of the time series data is additionally extracted in the process of distilling, which enhances the capacity of the model to capture long-range dependencies (Zhu et al., 2023a).

The distilled representation is expressed in Eq. (12):

$$X^{l+1} = MaxPool\left(Conv1D\left(X^l\right)\right) \quad (12)$$

where X^l is the representation at layer l.

Figure 4 shows the Informer model, a special version of Transformer that is efficient in long-sequence time-series prediction. The encoder takes the input time-series data on the left and converts it into embeddings by applying a 1D convolutional token embedding layer

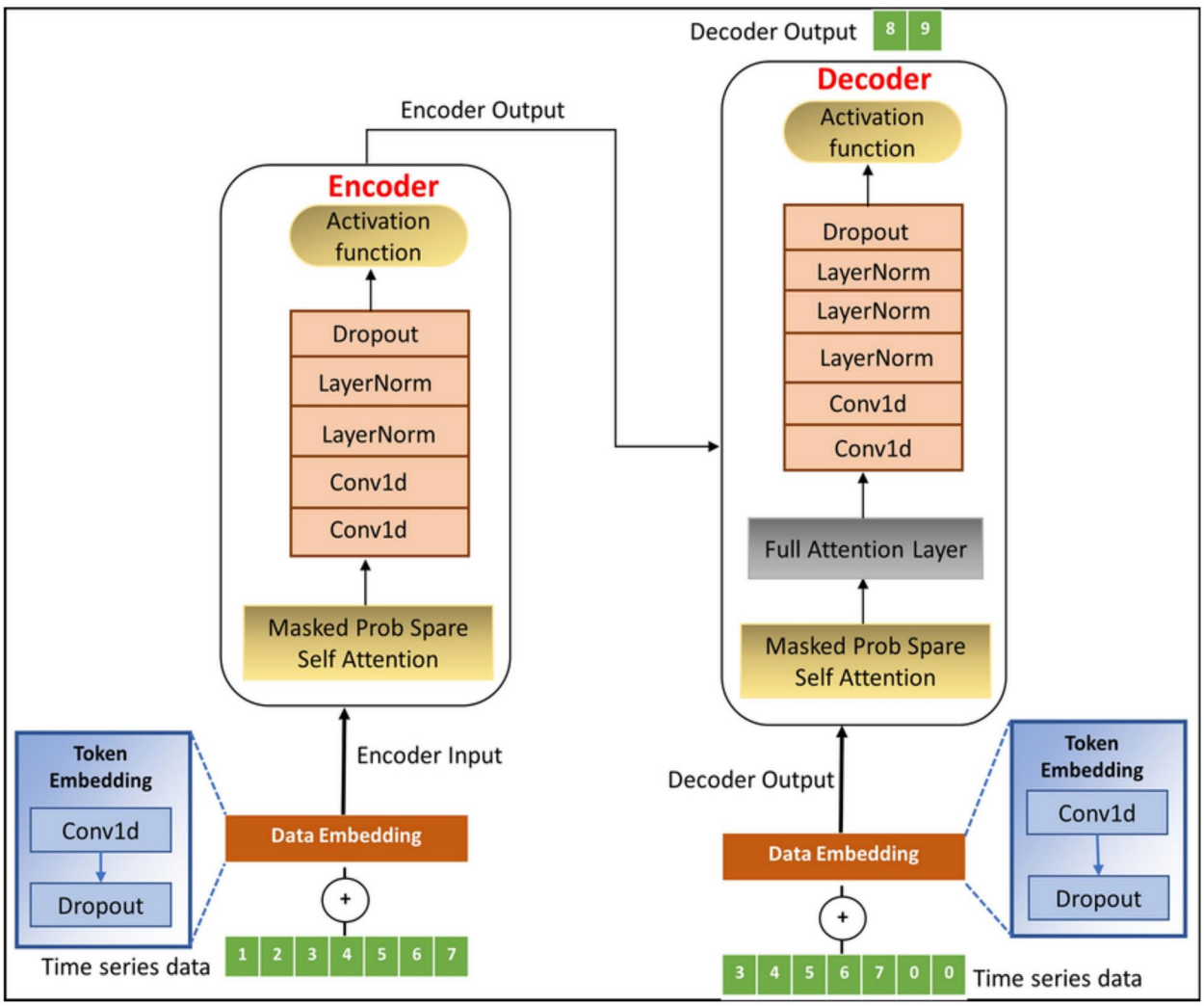


Fig. 4 Informer transformer architecture for time-series forecasting. (Source: Adapted from (Zhou et al., 2021))

plus dropout regularization. The embeddings are then fed into the encoder, with the most important operation being the Prob Sparse self-attention mechanism, a sparsity-based attention mechanism that needs substantially less computational power than a full-attention approach. Within the encoder, the representations are further refined by stacked layers of 1D convolutional filters, layer normalization, dropout, and activation functions to form representations that represent local and global dependencies (Zhou et al., 2021).

The right decoder is supplied with shifted time-series data, and the encoder outputs to predict the future sequences. Besides a complete attention layer to sufficiently integrate the information that can be found in the encoder, there is also the masking of Prob Sparse self-attention to avoid information leakage between time steps. Just like with the encoder, convolutional and normalization layers also exist with the decoder, which ensure steady learning. Finally, the decoder unfortunately, provides its estimation of the sequence based on its activation mechanism Mao et al. (Mao et al., 2024).

Overall, this type of architecture realizes efficiency and accuracy by a set of sparse attention, convolutional embedding layers, and an encoder-decoder architecture design, and can be particularly useful in learning long-range dependencies in time-series forecasting tasks.

Informer Transformer has been demonstrated to be superior to the standard transformer besides other time-traditional models in long-time series forecasting (Walczewski & Whrle, 2024). The long sequence processing power, as well as long-range dependencies, make it appropriate for solar PV power forecasting, but in this case, the historical data can be several years long (Song et al., 2024).

Temporal fusion transformer

The Temporal Fusion Transformer (TFT) is a transformer-based neural network that is explicitly trained to make predictions on time series with slow-varying dynamics and heterogeneous phenomena (Santos et al., 2022). It adopts various important attributes to enhance the accuracy and interpretability of forecasting through variable selection networks, temporal self-attention networks, and gating (Lim et al., 2021; Yang et al., 2024b). The Gate layer is implemented using Gated Linear Units (GLUs) as shown in Eq. (13), which help suppress less informative components of the network for a given dataset. The GLU operation is defined as:

$$GLU(X) = \sigma(W_1X + b_1) \odot (W_2X + b_2) \tag{13}$$

Add & Norm refers to the combination of residual connections and layer normalization as shown in Eq. (14)-(16), which has been shown to improve feature extraction in transformer-based architectures. The Gated Residual Network (GRN) further enables selective nonlinear processing. The structure of the GRN is given by:

$$GRN(a, c) = LayerNorm(a + GLU(h_2)) \tag{14}$$

$$h_1 = \mathrm{W}_3 h_2 + \mathrm{b}_3 \tag{15}$$

$$h_2 = \mathrm{ELU}\,(\mathrm{W}_4 \mathrm{a} + \mathrm{W}_5 \mathrm{c} + \mathrm{b}_4) \tag{16}$$

Variable selection Networks allow the TFT to choose automatically the best input features to predict (Islam et al., 2023). This can come in handy, especially when it comes to solar PV power prognostication, where any number of meteorological variables could affect production, including solar irradiance, temperature, and wind speed. Variable selection networks are trained to attribute weight to each input feature, such that they can only pay attention to the most important information and drop any irrelevant or noisy feature Islam et al. (Islam et al., 2023). As demonstrated in Fig. 5 (Ayhan et al., 2024), temporal self-attention layers recognize the temporal dependencies of the time series data in the same way as the standard transformer (Islam et al., 2023). But the self-attention layers in the TFT are augmented by gating schemes that regulate the information flow between time steps. These gating mechanisms enable the model to focus on the various time steps depending on their importance to the task at hand when forecasting (Mouloud et al., 2023). The conventional multi-head attention mechanism is defined in Eq. (17):

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{17}$$

where Q, K, and V denote the query, key, and value matrices, respectively, and d_k is the dimensionality of the key vectors. A masking matrix M is introduced as shown in Eq. (18) to guide the flow of information and improve interpretability, leading to the masked attention formulation:

$$MaskedAttention(Q, K, V, M) = softmax\left(\frac{QK^T + M}{\sqrt{d_k}}\right)V \tag{18}$$

Masking matrix M induces large negative values in masked locations such that softmax output of these locations is closer to zero, causing attention to be drawn toward the important features. The interpretable multi-head attention further applies rectification to maintain non-negative attention scores, as shown in Eq. (19) given by:

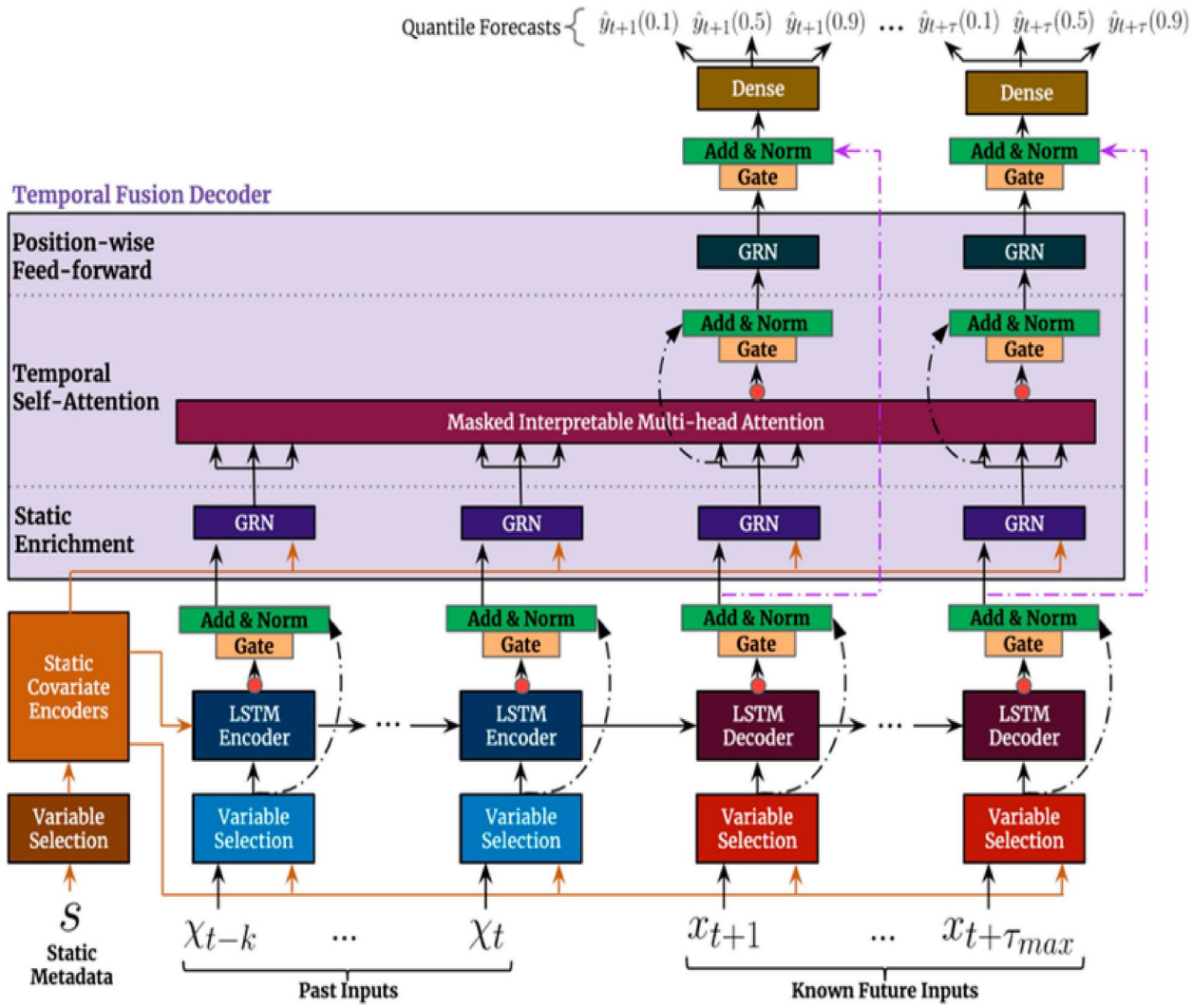


Fig. 5 Top-level architecture of TFT, along with its main components. (Source: Adapted from (Mouloud et al., 2023)

$$RectifiedAttention(Q,K,V,M) = relu(MaskedAttention(Q,K,V,M)) \quad (19)$$

Analysis of model behaviour using a combination of masking and rectification leads to a greater interpretability of the attention mechanism. Transparency in model development is particularly important in applications such as solar PV performance modeling, where trust in predictions is required. The TFT architecture, supported by the above mathematical formulation, enables the extraction of complex temporal relationships in multivariate time-series data and is therefore suitable for solar energy performance modeling tasks.

To overcome the heterogeneous data, the TFT also includes the static covariate encoders and the known future input encoder (Mouloud et al., 2023). Time-invariant covariate encoders (location and installation) can be handled through the future input encoder (known future), and time-varying covariates (planned grid outage or scheduled maintenance) can be handled with time-invariant covariate encoders (location and installation).

TFT is capable of processing large amounts of heterogeneous data with complex temporal dynamics, which is why it is a good fit to solve solar PV power predictions, Santos et al. (Santos et al., 2022). It has variable selection networks, temporal self-attention layers, and gating mechanisms that allow it to learn how meteorological variables, historical meteorological PV power data, and other relevant variables interact in a complex manner (Mouloud et al., 2023). Moreover, its readable architecture enables users to learn the aspects underpinning the predictions they make through the TFT, which can

be helpful in making decisions Islam et al. (Islam et al., 2023).

Hybrid transformer model

Hybrid transformer models result from integrating the benefits of transformer architectures with other deep learning models or statistical approaches to refine the performance of forecasts (Mao et al., 2024). The purpose of these hybrid models is to overcome the weaknesses of individual models and use the advantages of each one to obtain a more accurate and robust prediction (Mao et al., 2024). There are models of hybrid transformers which are suggested to be used in solar PV power forecasts, such as CNN-Transformer, LSTM-Transformer, and GRU-Transformer models (Ibrahim et al., 2024) (Salman et al., 2024).

The CNN-Transformer models are another type of model that integrates both convolutional neural networks (CNNs) and transformer architectures to learn spatial and temporal features of solar power data (PV) (Khan et al., 2024). CNNs are typically good at finding spatial information in images or time series data, whereas transformers are good at finding long-range temporal dependencies (Wang & Kang, 2024). The integration of vision-based Transformer components like Swin Transformer (Liu, 2021) and ViT (Dosovitskiy, 2021) has further improved the processing of spatial weather tensors. A combination of these two architectures helps CNN-Transformer models to effectively learn the multifaceted spatio-temporal distributions of solar PV power measurements and thus achieve better forecasting performance (Khan et al., 2024).

LSTM-Transformer models are LSTM-based networks paired with transformer-based networks that take advantage of the benefits of both model types (Jailani et al., 2023). LSTMs are best suited for processing sequential data and learning short-term temporal dependencies, whereas transformers excel at learning long-term dependencies (Unlu et al., 2022). The integration of these two architectures allows LSTM-Transformer models to effectively introduce short and long-term variation in solar PV power data and make more accurate predictions (Mao et al., 2024).

The decision as to what hybrid transformer model to apply lies with the unique features of the solar PV power data and the forecasting mission, Mao et al. (Mao et al., 2024). Figure 6 demonstrates CNN-Transformer models as well adapted to capture spatio-temporal patterns and LSTM-Transformer or GRU-Transformer as well adapted to capture both short and long-term temporal dependencies (Khan et al., 2024) (Al-Ali et al., 2023). Depending on the way the required hybrid architecture is selected, one

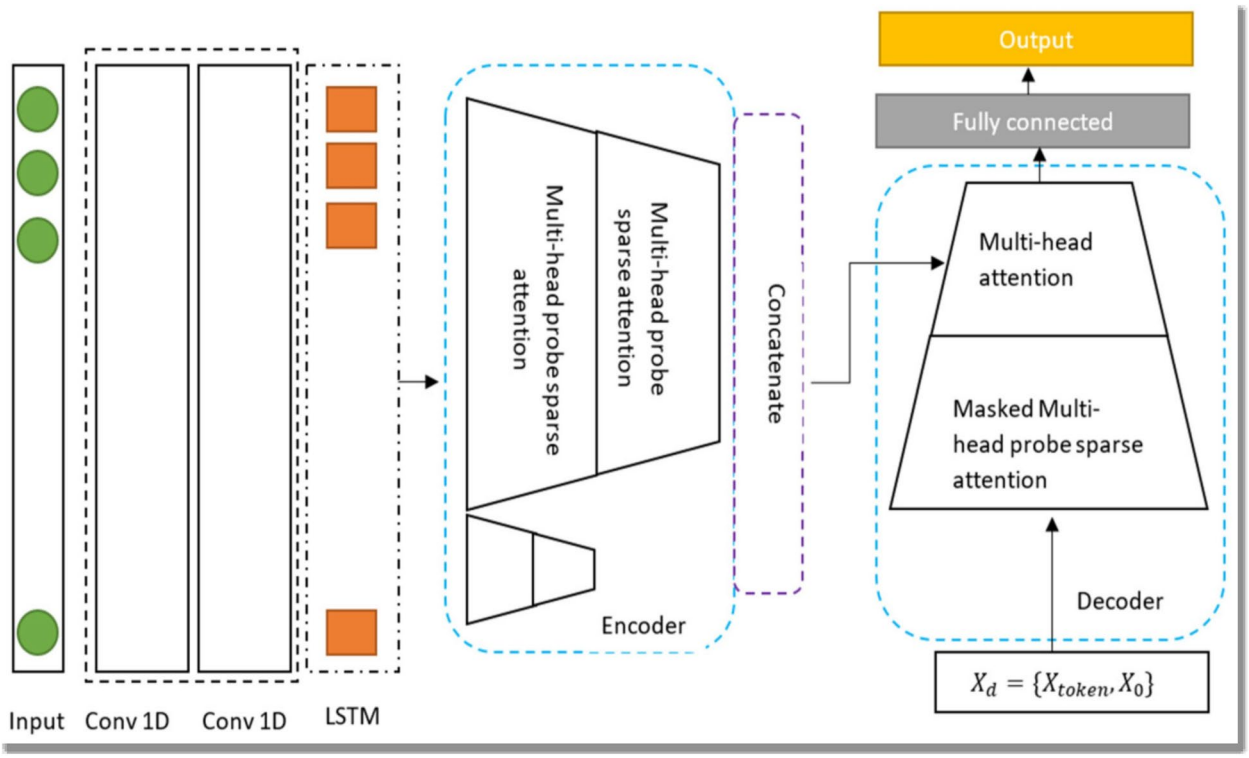


Fig. 6 Structure of hybrid CNN-LSTM-transformer model. (Source: Adapted from (Ibrahim et al., 2024))

can enable considerable gains in solar PV power forecasting accuracy (Mao et al., 2024).

GRU-Transformer models resemble the LSTM-Transformer models, only that they couple LSTMs with gated recurrent units (GRUs) (Mao et al., 2024). GRUs are simplistic variants of LSTMs with fewer parameters and, in general, are easier to train Wu et al. (Wu et al., 2024). GRU-Transformer models can reach similar performances to LSTM-Transformer models with lower interventional cost, and thus become a feasible choice to predict solar PV power (Mao et al., 2024).

Hybrid GRU-transformer model

This model uses a dual-layer architecture to uphold the accuracy of prediction (Mao et al., 2024). The former employs GRUs to condense the most important temporal statistics of the stream of data, and the latter employs the self-attention mechanism of the Transformer in discovering global facts and temporal situational dependencies in the stream of temporal data (Mao et al., 2024). This cleans up the sequence data and increases the representational ability of the model, and this leads to the accurate short-term predictions of the PV power. It is empirically confirmed that the GRU-Transformer model works better than all existing methods in the accuracy of predictions made using actual measurement data, (Mao et al., 2024).

The Hybrid GRU-Transformer model addresses the dual challenge of short-term volatility and long-term trends in PV forecasting. As shown in the new Fig. 7, the model has a parallel architecture. GRU would identify the important short-term temporal statistics through historical sequences (like sudden drops in power caused by moves of clouds) (Mao et al., 2024). At the same time, the Transformer branch exploits a self-attention (discovering global facts and situational dependencies), e.g., the dependence between the current irradiance stages and seasonal production standards (Mao et al., 2024). These qualities can be combined to enable the meta-learner to clean up the received sequences and enhance the representational capacity of the model which results in extremely accurate short-term (15 min to hourly) predictions.

Structure of the hybrid GRU-transformer for solar PV

The model will start with correlation analysis to detect the important factors affecting PV generation, including solar irradiance and ambient temperature and after that, they will be combined to create a single embedding. The architecture establishes the use of two branches as upholders of prediction accuracy:

- Input Layer: Processes multivariate time-series data, including historical PV output, solar irradiance (GHI), and ambient temperature.
- GRU Branch: This uses Gated Recurrent Units to compress and mining important short-term temporal data of the data stream (e.g., fast changes in output due to movements of clouds).
- Transformer Branch: This parameter is based on the self-attention mechanism, to find out global tempo-

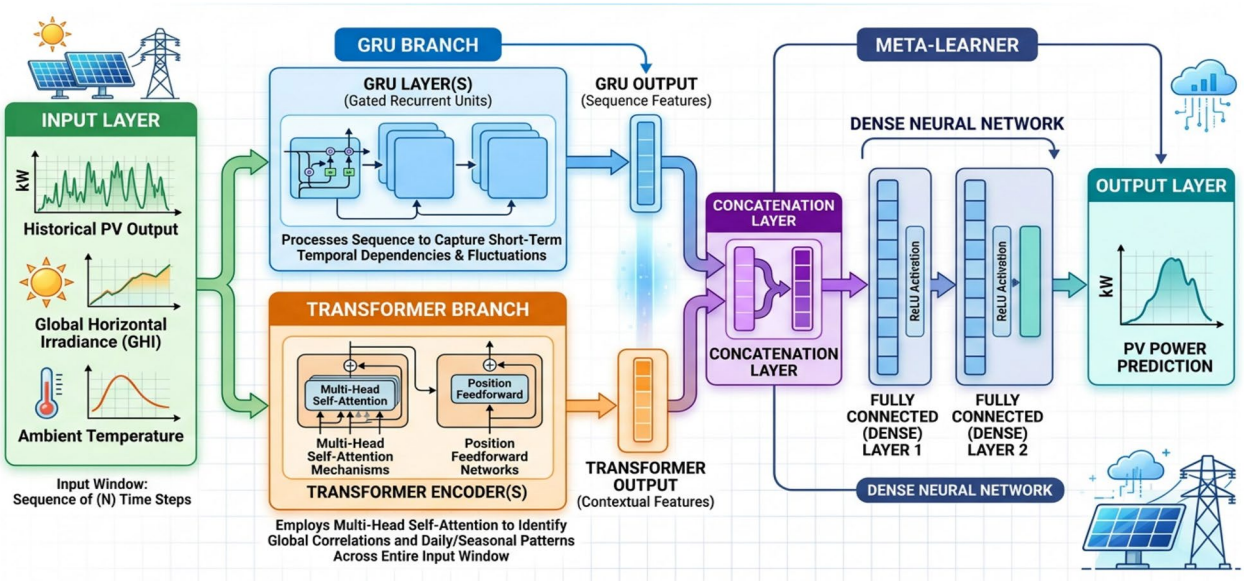


Fig. 7 Hybrid GRU-transformer framework for short-term solar PV power prediction. (Source: created by the author using Napkin AI.)

ral relationships, and seasonal patterns that recurrent units may overlook.

- Fusion and Prediction: The extracted features from both branches are concatenated and passed through a meta-learner (typically a series of dense layers) to generate the final short-term power forecast.

Patch-transformer model

The Patch-Transformer model is a new model that is designed to enhance scalability, perception, and generalisation capacity of the PV power prediction model by applying a combination of the Patch and Transformer method (Zhan et al., 2024). The problems that are prevalent in older versions of neural networks include the variable power production of PV, thus the forecasting and control of power grid instability (Zhan et al., 2024). Patch-Transformer addresses this by chunking the input data as a patch, such as images in computer vision, in such a way that the model focuses on local patterns and relationships in the time series (Huo et al., 2025).

Patch processing helps the model to reduce the computational complexity of long time series and simplify it to be more efficient and scalable (Zhan et al., 2024). Transformer architecture subsequently acquires global dependencies between these patches and enables the model to think about the high-level picture and make better predictions. This scheme is already proven to be sound during various seasons and thus has the potential of replacing grid reliability in the renewable energy industry Zhan et al. (Zhan et al., 2024).

Decomposable transformer (TSformer)

The Decomposable Transformer, or TSformer, is designed to model trend and seasonal components of time series independently according to their varying properties (Hu et al., 2024). TSformer, inspired by PatchTST and iTransformer, is a Transformer encoder-only architecture that can capture the inter-series relations and intra-series time dynamics quite well (Hu et al., 2024). The weather situation exposes PV power generation to intermittency, volatility, and uncertainty, resulting in significant errors in traditional prediction techniques (Hu et al., 2024).

To avoid such challenges, TSformer breaks down the time series into trend and seasonal components so that the model can concentrate on the additional features of each (Hu et al., 2024). The trend component is based on the long-term trend of the series, and the seasonal component is based on the repetitive trends within the data. Through the component-based modeling, TSformer can yield better prediction ability and stronger robustness than other baseline models, thus can serve well in photovoltaic power forecasting Hu et al. (Hu et al., 2024).

Weather embedding cluster transformer (WECT)

As shown in Fig. 8, the Weather Embedding Cluster Transformer (WECT) model combines weather clustering and similar-day clustering methods to enhance the solar irradiance forecasting (Wang & Kang, 2024). The current models do not usually consider the effects of climate situations on solar radiance and cannot account for multi-scale data characteristics (Wang & Kang, 2024). WECT solves this problem by initially classifying weather based on the K-means clustering algorithm, applied to observational data, including ground-level relative humidity, ambient temperature, and solar irradiance Wang & Kang (Wang & Kang, 2024).

Pre-processing is then done on historical solar irradiance data into 12 h slices with both the weather and the locality information embedded (Wang & Kang, 2024). Two Transformer modules are developed: a Temporal Transformer module and a Similar-Day Semantic Transformer module (that is capable of capturing long- and short-term temporal features and solar irradiance patterns, respectively) (Wang & Kang, 2024). Self-attention mechanism in these modules determines the most important frames to make predictions and takes more weight in the neural network. Several experiments on actual datasets of the populace propose that WECT is a dependable forecast of solar irradiance, Wang & Kang (Wang & Kang, 2024).

CT-PatchTST (channel-time patch time-series transformer)

Channel-Time Patch Time-Series Transformer (CT-PatchTST) is a multidimensional neural network, created to predict the amount of power produced by photovoltaic power plants and wind energy plants (Huo et al., 2025). The initial version of the Patch Time-Series Transformer (PatchTST) employs the channel-independent (CI) approach, where inter-channel interaction is automatically ignored throughout the learning process and may result in the loss of relevant information (Huo et al., 2025). To compensate for this disadvantage and increase the opportunities for the finer granularity of the data represented by CI, CT-PatchTST adds to the processing of inter-channel data, retaining the advantages of the channel-independent strategy (Huo et al., 2025; Nie et al., 2023).

The improved version of this is strictly learned to calculate the fact that it can precisely and reliably forecast energy (Huo et al., 2025). The CT-PatchTST will be advantageous in the sense that it will be able to capture increasingly more detailed information through the examination of interactions of different channels (e.g.,

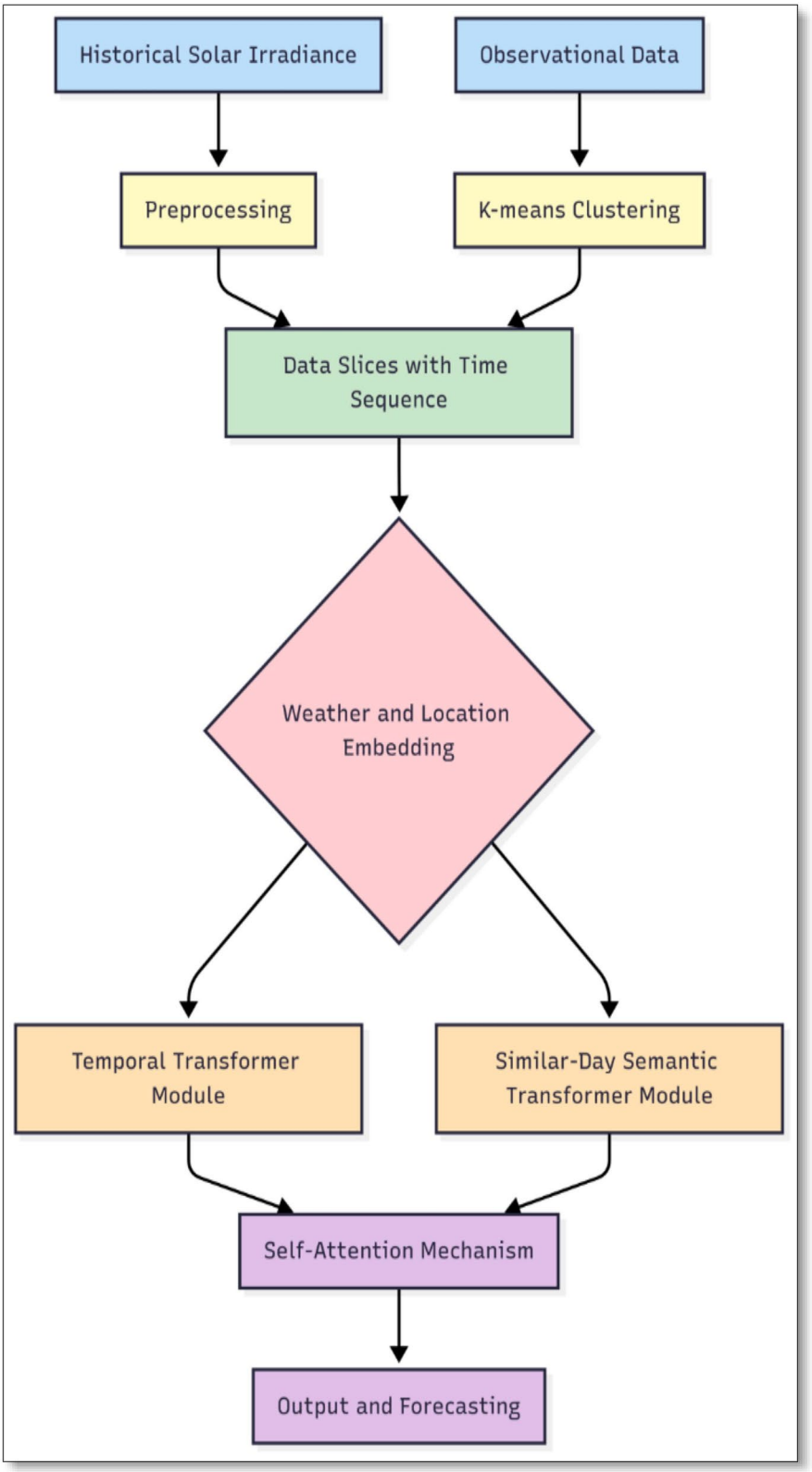


Fig. 8 Structure of weather embedding cluster transformer (WECT) for solar irradiance forecasting. (Source: Created by the author using Napkin AI.)

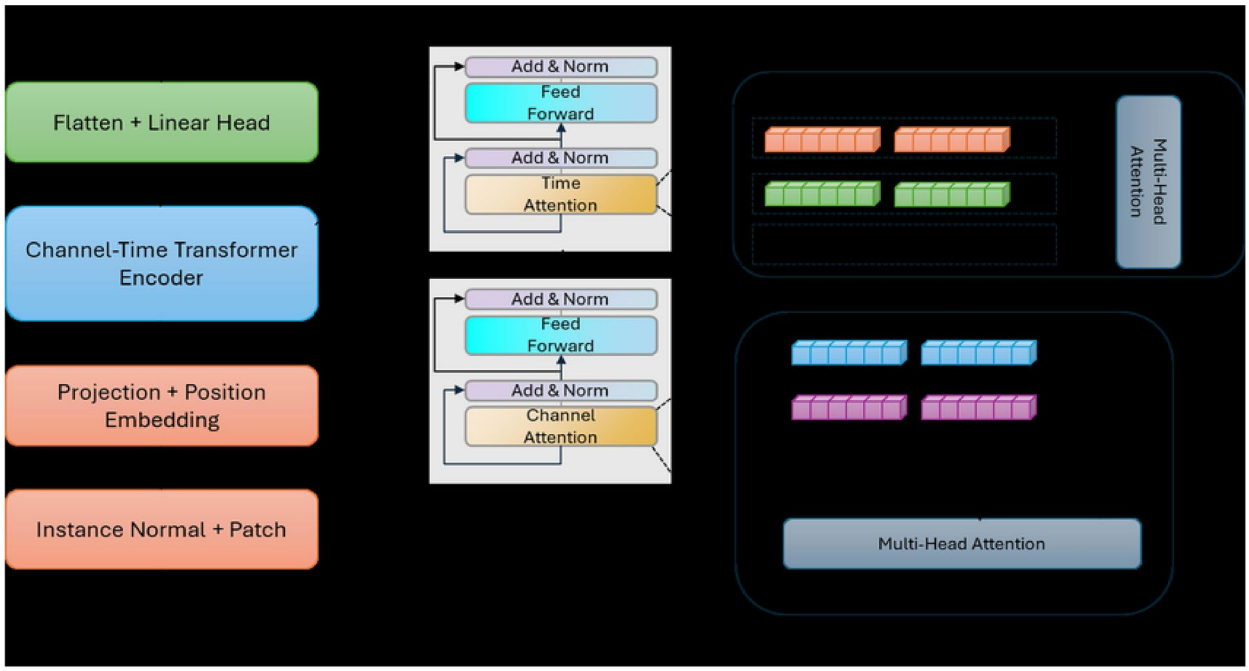


Fig. 9 Schematic architecture of channel-time patch time-series transformer framework for sequence modeling. (Source: Adapted from (Huo et al., 2025))

different sensors or sources of data) to increase the accuracy of long-term renewable energy forecasting, Huo et al. (Huo et al., 2025).

Figure 9 (Huo et al., 2025) illustrates the architecture of a Channel-Time Transformer model that is formed to learn sequences. The second stage at the bottom then normalizes the data distribution of the input, which is then followed by the Patch extraction, which divides the input into small patches. An individual patch is then consecutively sent to a Projection and Positional Embedding layer to capture the time order. The data is then converted by the Channel-Time Transformer Encoder, in which two significant attention mechanisms include Time Attention that models dependencies between the time steps, and includes Channel Attention that models dependencies between the various feature channels. Each block of attention consists of normalization, feed-forward layers, and multi-head attention that adds to the learning of a representation. The final phase is flattening the encoded features and directing them into a linear head to accomplish downstream projects such as prediction or classification. It has been demonstrated that the architecture brings a pleasant combination of dynamical semantics of time and feature association, and thus the structure has been applicable in a broad area of applications, which include forecasting time series and analysing multivariate signals.

Multi-branch ResNet-transformer (ResTrans)

Multi-Branch ResNet-Transformer (ResTrans) architecture Residual network Multi-Branch ResTrans is a spatial–temporal model that is a hybrid of the Residual Transformer architecture and the Transformer architecture (Ziyabari et al., 2023). The current techniques are likely to face setbacks in both the spatial covariance between the adjacent solar generation areas, the examination of the similarity of patterns for the long-term and changing with time, and the mitigation of overtraining in convolutional and recurrent neural networks (Ziyabari et al., 2023). To face these challenges successfully, it has been proposed that ResTrans be suggested (Ziyabari et al., 2023).

It has been evaluated using real-life data consisting of 17 years of data of various solar loci with the greatest forecasting resolution of any benchmark hybridizing architecture (Ziyabari et al., 2023). The multi-branch architecture enables the model to simultaneously represent the various attributes of the spatio-temporal data, and the ResNet and Transformer blocks support the successful extractions of features and long-range dependency relationships (Ziyabari et al., 2023). This renders ResTrans a credible short-term prediction of solar irradiance forecasting, Ziyabari et al. (Ziyabari et al., 2023).

SL-transformer

The SL-Transformer is an innovative framework that relates to the deep learning framework but makes adjustments to the specifics of green forecasting, requiring accurate forecasts of the produced power by wind and solar energy sources (Zhang et al., 2023; Zhu et al., 2023b). Temporal complexities and noise are thorny to the existing models because they are large and require more power (Zhu et al., 2023b). To be able to have a stable base on which to proceed with the analysis and modeling, free of outliers, the SG filter and LOF algorithm are applied in data cleansing (Zhu et al., 2023b). Moreover, a self-attention mechanism is added to enable the model to be more capable of detecting dynamically and fast adjusting the input weights (Zhu et al., 2023b). The SL-Transformer obviously outperforms other prime models benchmarked in terms of these two hallmarks to have the near-perfect value of R^2 equal to 0.9989 and immensely low SMAPE error at 5.8507% in prediction (Zhu et al., 2023b). To forecast, it has been found to have forestalled 4.2156, which is a decent 15 percent gain compared to the competing prediction models. The effectiveness of it is demonstrated by the outcomes of the experiment Zhu et al. (Zhu et al., 2023b).

To deliver a better overview, Table 2 summarizes the considered transformer variants with their respective goals, use rate, input type of data, and the AI methods used to make a solar PV forecast.

Unified benchmarking framework

One key limitation found during this review is that there is no standardized comparison of Transformer variants as they are based on different datasets and experimental conditions. To address this, we are suggesting a common empirical template which aims at shifting the field to qualitative descriptions to a rigorous quantitative assessment.

Standardized dataset selection

We identify a "Gold Standard" publicly available dataset that become the key milestones of testing the future Transformer architecture in solar PV:

The dataset is that of a utility-scale solar PV plant in Vadodara, Gujarat, India. It is designed with a rated capacity of about 10 MW and comprises crystalline silicon PV modules that are interconnected by a series of inverters. This system data were measured in three years (20,152,017) and covers each day, AC energy production and environmental variables (solar irradiance, module

Table 2 Summary of transformer variants and their applications

References	Year	Transformer variant	Focus on	Application used	Type of data	AI techniques
(Vaswani 2017)	2017	Simple Transformer	Baseline forecasting	Short-term PV power prediction	Time series data	Self-attention, encoder-decoder
(Santos et al., 2022)	2022	Temporal Fusion Transformer (TFT)	Feature selection, interpretability	Day-ahead PV power forecasting	Time series and meteorological	Deep learning, attention, and gating
(Islam et al., 2023)	2023	TFT	Multivariate forecasting	PV plant output prediction	Time series + numerical	Variable selection networks
(Zhan et al., 2024)	2024	Patch-Transformer	Scalability, local–global learning	Short-term PV power prediction	Time series patches	Hybrid deep learning
(Hu et al., 2024)	2024	Decomposable Transformer (TSformer)	Trend-seasonality decomposition	Multi-horizon PV power prediction	Time series and NWP data	Deep learning, series decomposition
(Wang & Kang, 2024)	2024	WECT	Weather-based clustering	Solar irradiance prediction	Numerical and weather clusters	Clustering and transformers
(Huo et al., 2025)	2025	CT-PatchTST	Channel-wise dependency modeling	Long-term PV forecasting	Multivariate time series	Attention and patch embeddings
(Mao et al., 2024)	2024	GRU-Transformer	Short-term dependency modeling	Hourly PV forecasting	Time series and meteorological	Hybrid RNN and Transformer
(Ziyabari et al., 2023)	2023	ResNet-Transformer (ResTrans)	Spatio-temporal forecasting	Multi-site solar irradiance forecasting	Time series and image	CNN and Transformer
(Zhu et al., 2023b)	2023	SL-Transformer	Noise removal & enhanced accuracy	Wind and PV forecasting	Time series and SG filter	LOF and Transformer
(Wu et al., 2024)	2024	Hybrid Transformer	Weather-classification forecasting	Day-ahead PV prediction	Weather and numerical	Hybrid DL and clustering
This Review	2025	All Transformer Variants	Comparative study	PV power forecasting	Time series, NWP, and image	Deep learning and hybrid architectures

temperature). The dataset is split into two major subsets namely training and testing. For the training, the 2015 and 2016 dataset (731 samples, 70% of the total data), is utilized, and remaining 30% (304 Samples) is used as the validation.

Controlled experimental protocol

To make our studies reproducible, we come up with a standardized protocol to be used in future comparative studies. The following controlled variables should be considered in evaluating Transformer, Informer and TFT models:

We have made a parametric comparison of the various deep learning architectures, including Transformer, Informer and TFT in this work. PyTorch was used as the deep learning framework to train and test the models on a system with an Intel(R) Xeon(R) E-2136 CPU, 3.30 GHz (3.31 GHz) and 16.0 GB RAM.

We experimentally tuned several important parameters to examine the impact of different hyperparameters on the performance of the model. The learning rate (LR) indicating the size of the weight updates was experimented between the magnitude of 1e-5 and 1e-2 with the decay being exponential when needed. The trade-off between the training stability and memory efficiency was optimized by investigating the batch sizes, 16, 32, 64 and 128. Layers were set to numbers of 2 to 8 encoder/decoder layers that were assessed. The number of neurons per layer, which will be denoted by the name of the hidden units, was between 64 and 512, and was chosen according to the constraints of the computations. Dropout rates of 0.1, 0.2, 0.3 and 0.5 were used to minimize overfitting. To analyse the effect of multi-head attention in Transformer-based models, attention heads were varied in number (4 to 16) to study their effects on the model. The length of the sequence or the temporal window of input data was varied with 10, 25, 50, and 100-time steps according to the nature of the data. Finally, various optimizers such as Adam, RMSprop and SGD were tried with various momentum

Table 3 Comparative performance metrics of selected transformer variants (Sharma 2025)

Model	MAE	MSE	RMSE	Result summary
Transformer model	1.55	0.05	2.18	Baseline temporal dependency tracking
Informer transformer	1.42	0.05	2.26	Improved computational efficiency and stable long-sequence forecasting through sparse attention
Temporal fusion transformer	0.95	0.02	1.44	Best-in-class precision and gating

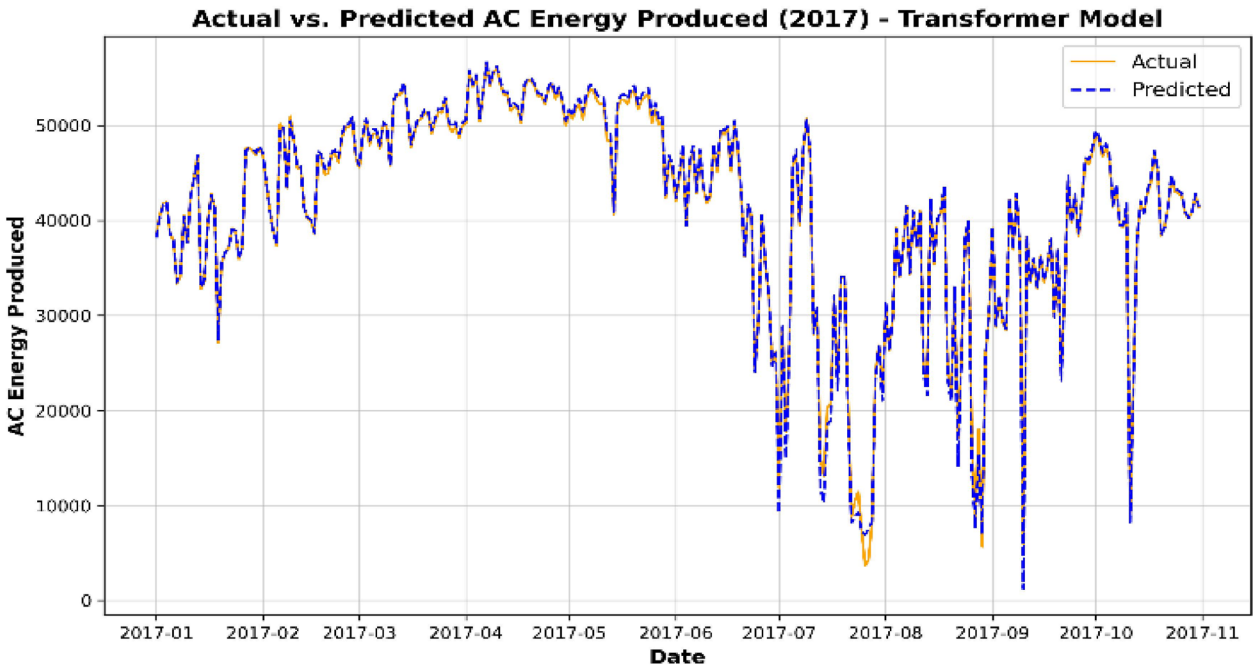


Fig. 10 Actual vs. Predicted AC Energy Produced (2017) – Transformer Model. (Source: Adapted from (Sharma 2025))

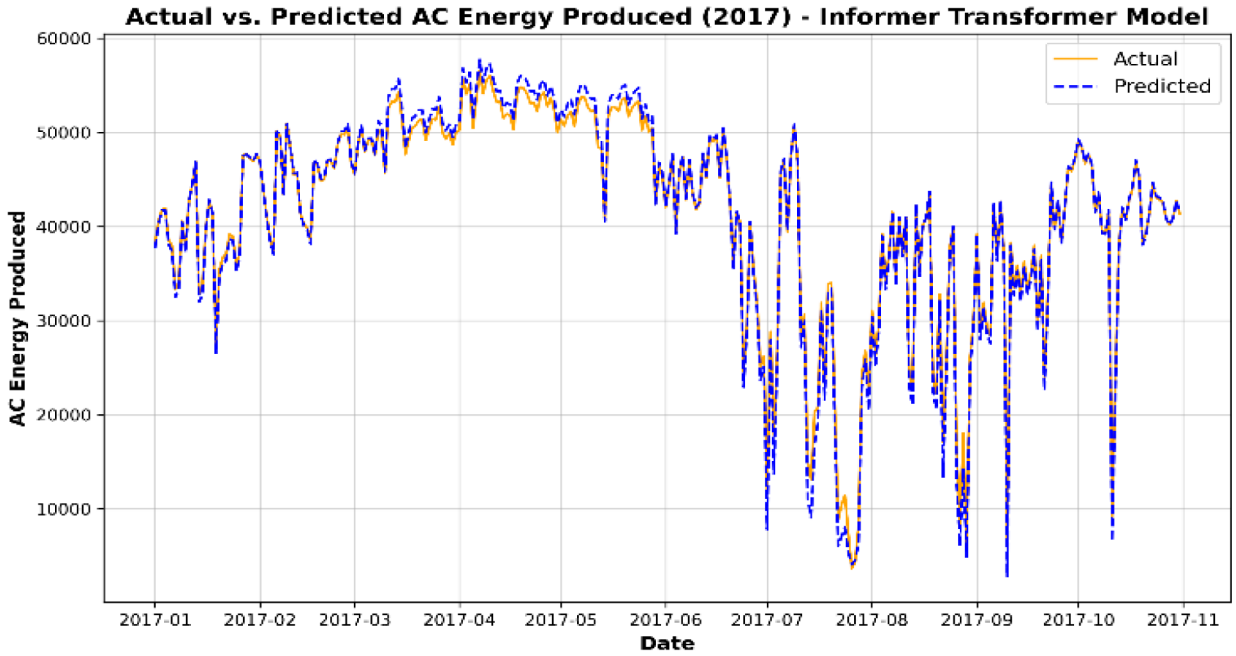


Fig. 11 Actual vs. Predicted AC Energy Produced (2017) – Informer Transformer Model. (Source: Adapted from Sharma (2025)

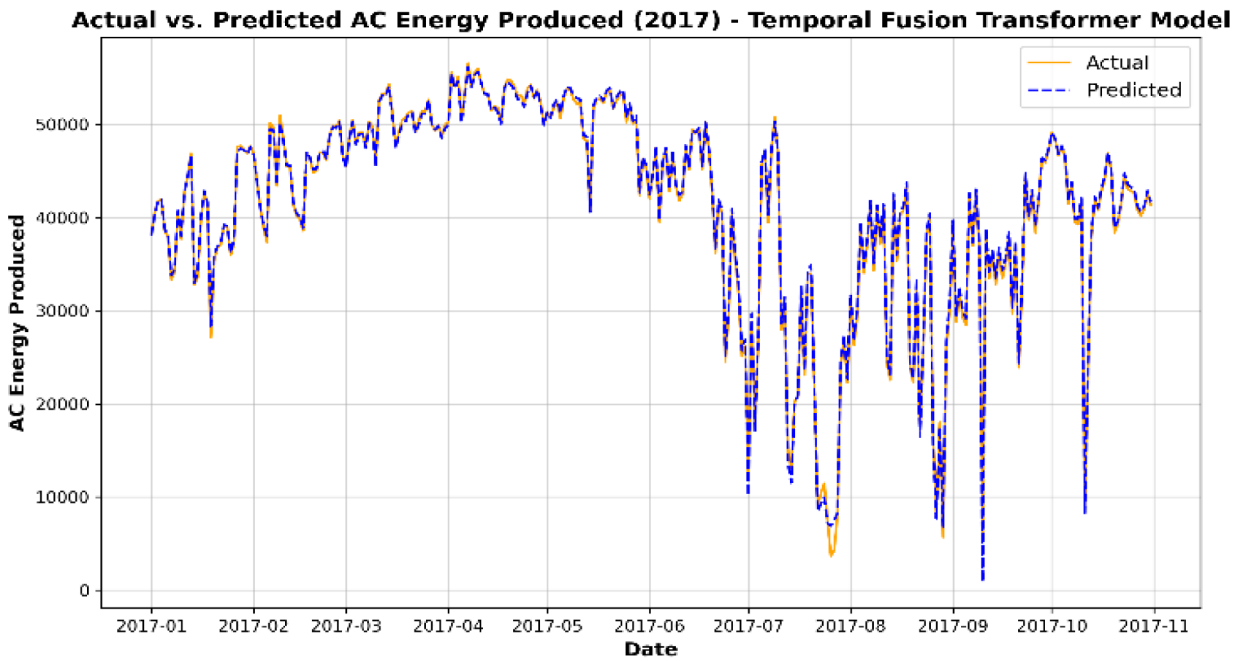


Fig. 12 Actual vs. Predicted AC Energy Produced (2017) – Temporal Fusion Transformer Model. (Source: Adapted from Sharma (2025)

and weight decay to determine their effect on convergence and stability of training. A range of assessment measures was applied to test the performance of the various models and they included RMSE, MSE and MAE (Sharma 2025).

Standardized formative assessment on data

To prove our theoretical framework and overcome the shortage of the qualitative comparison, we provide the preliminary empirical inquiry of three exemplary versions of Transformers Standard Transformer, Informer, and TFT. These models are chosen because they are the development of attention mechanisms through the history of standard self-attention to dedicated time-series architectures (Ahmed et al., 2025).

Quantitative results and analysis

To add some qualitative aspect to the quantitative performance measures in Table 3 (Sharma 2025), we show the visual tracking performance of the main Transformer variants. Figure 10, 11, 12 display actual and predicted AC energy generation of the base Transformer, Informer as well as the Temporal Fusion Transformer models.

We can visualize the response of each architecture to the stochasticity of solar data-ranging, including the basic temporal mapping of the basic Transformer, to the high-density, gated alignment of the Temporal Fusion Transformer (Roy et al., 2024) by visualizing these prediction trajectories across a representative sample period (Sharma 2025).

The table below represents the brief analysis of the performance parameters Mean Absolute Error (MAE), Mean Squared Error (MSE) and Root mean Square Error (RMSE) of the evaluated architectures in regard to the standardized dataset (Sharma 2025).

Analysis of comparative results

The relevance of the prediction trajectories of the three variants visually verifies their comparative performance. Figure 10 demonstrates that the standard Transformer Model is able to follow the general diurnal cycle, and thus, reflects the key temporal correlations of the PV system. Figure 11 shows the Informer Transformer Model exhibiting stable forecasting behavior and efficient long-sequence modeling through its ProbSparse attention mechanism. While its RMSE is slightly higher than that of the standard Transformer on the selected benchmark dataset, Informer offers significant computational advantages for large-scale forecasting applications. Lastly, Fig. 12 demonstrates the broadest capacity of the Temporal Fusion Transformer that shows the maximum level of agreement with the ground-truth AC energy production even in the high volatility. This visual outcome together with the error improvements in Table 3 are the needed validation that can justify our architectural statements that are well controlled.

Limitation of benchmarking framework

Interpretation of the benchmarking results for this review must be interpreted with caution. The comparative assessment was carried out with one solar PV data set from a utility scale PV system at Vadodara, India, and analyzed three representative transformer architectures: Standard Transformer, Informer and Temporal Fusion Transformer. These models were chosen as they span various phases in the development of attention-based forecasting models but results may not be applicable to other climatic regions, PV technologies, data resolution or forecasting timeframes. Additionally, some newly proposed transformer variants (e.g., Autoformer, TSformer, CT-PatchTST, and WECT) were not practically tested because of the lack of a public benchmark dataset and variations in implementation across different studies. The benchmarking framework should be expanded to include additional geographically diverse datasets and a wider range of transformer architectures to make more robust and general conclusions in the future.

Operational and computational trade-off analysis

Although the accuracy measures are important, an analytical comparison should include consideration of the operating costs of each architecture. The trade-off between predictive precision and computational overhead is apparent as shown in Table 4. Although a Temporal Fusion Transformer (TFT) is the most accurate, its strong performance demands extensive feature engineering and cannot be implemented as easily because of its complicated gating processes. On the other hand, the Informer and Autoformer versions provide a more desirable trade-off towards real time usage because it provides reduced computational complexity

Table 4 Operational and computational comparison of primary transformer variants

Model	Data requirement	Training time	Computational cost	Implementation difficulty
Standard Transformer	Medium	High	High $(O(L^2))$	Low (Standard Libraries)
Informer	High (Long History)	Low	Low $(O\ (L\ log\ L))$	Moderate
TFT	High (Multivariate)	Moderate	Moderate	High (Complex Gating)
Autoformer	High	Low	Low $(O\ (L\ log\ L))$	Moderate

compared to $O\ (L^2)$ and one of $O\ (L\ log\ L)$ training of long-sequence historical data.

Practical implementation and operational challenges

Transformer architectures are more predictive but have serious application issues in real world PV systems. Practically, SCADA systems often experience communication delays and lost data packets, the latter may interfere with the fixed-window needs of attention systems. In addition, sensor errors and drifting calibration in pyranometers bring about soft noise, which has the potential to produce model divergence (Sharma et al., 2024) (Sharma et al., 2023) (Malik & Chandel, 2020). As preventive measures to these, we would suggest that a formidable preprocessing layer should be integrated e.g. the SG Filter and the LOF algorithm in the next Sect. 4.11 which has been demonstrated to offer a near-perfect R^2 of 0.998 even in adverse noise conditions. Lastly high cost of computation to train these models requires an apparent 'Model Update Requirement' schedule; retraining every month with transfer learning is suggested to keep accuracy reasonable and without excessive overload on GPUs in highly non-stationary climates (Wang et al., 2024).

Discussion

Lastly, the popularity of applying these highly effective deep learning-based transformer models to predict solar PV power can be seen by the broad range of transformer architectures introduced in this review. Each of them possesses its benefits and solves part of the time-series forecasting problems, like the long sequences, non-stationary data, and interpretability (Hertel et al., 2022) (Lee & Jeong, 2025) (Shih et al., 2023). The various transformers and their applications are already summarized in Table 2.

The Informer Transformer's main advantage for large scale and long sequence is attributed to its ProbSparse attention mechanism. The model also can selectively focus on the most informative queries, thereby significantly reducing the computational load compared to the use of the conventional self-attention mechanism, while still retaining the capability to capture long-range temporal dependencies. In addition, this mechanism can help filter out redundant information and enhance the scalability of the processing of large solar time-series. The Informer outperforms the standard Transformer in terms of RMSE on the benchmark set presented in Table 3, but its lower computational complexity and applicability for long-horizon forecasting make it a promising option for large-scale solar PV forecasting applications. So, the key benefit of Informer is not that it consistently outperforms all the datasets, but that it is computationally efficient and scalable.

Likewise, the quantitative improvements on Temporal Fusion Transformer (TFT) with its low MAE of 0.95 in particular can be readily traced to its Variable Selection Networks. In contrast to classic LSTMs where everything is fed the same the gating capabilities of the TFT enable it to effectively shut down less important meteorological parameters, e.g., humidity or wind speed, when clear skies prevail such that the major factor is solar irradiance only. This bias processing smaller feature space and avoids the dilution of important signals, and that is why the TFT demonstrates a better fusion with ground-truth generation in volatile weather conditions.

The summarization of check of datasets, sampling rate, forecast horizons, performance measures, and the key advantages of any model among a variety of transformer models are presented in Table 5, in comparison to others. The issue that must be tackled further research to improve the precision, durability and readability of the forecasting models using transformers to anticipate solar PV power generation is more convenient (Massaoudi

Table 5 Comparative Performance of Transformer Variants with Contextual Data

Transformer Variant	Dataset Source/Scale	Sampling Rate	Forecast Horizon	Primary Metric	Result
Simple Transformer	Benchmark datasets	Varied	Short-term	RMSE, MAE	Baseline
TFT (Santos et al.)	Real PV plant data	Hourly	Day-ahead	R^2, SMAPE	$R^2 = 0.92$
TFT (Islam et al.)	SCADA PV dataset	15-min	24 h	MAE	↓15%
Patch-Transformer	PV site data	15-min	Short-term	MAE	↓18%
TSformer	Multi-horizon sets	1-h	1–24 h	RMSE	↓22%
CT-PatchTST	Renewable datasets	Varied	1 week ahead	MAPE	↓20%
GRU-Transformer	SCADA PV data	15-min	15-min ahead	RMSE	↓12%
ResTrans	17-year solar dataset	Daily/Hourly	Multi-site	R^2	$R^2 = 0.96$
SL-Transformer	Wind and PV dataset	Hourly	24 h	R^2, SMAPE	$R^2 = 0.9989$

et al., 2021). This would involve creating new more interpretable transformer architectures, which would better estimate uncertainty, and attempting to find more hybrid approaches to combine transformers with other approaches to create more effective forecasts. Furthermore, research on how to formulate strategies to address missing data, outliers and other data quality issues that are likely to affect the quality of transformer-based forecasts must also be done (Phan et al., 2024). Measurements of performance as in Table 5 are to be treated within the context of their respective experiments. An example is that the SL-Transformer has near-perfect R^2 value of 0.9989, and much of this can be explained by the fact that it is very time aggressive in terms of noise-resilience using SG filtering and LOF outliers due to the outlier detection mechanism. In contrast, models such as the TFT have lower R^2 values of 0.92 but have more interpretability of nonhomogeneous variables like cloud cover and humidity.

The variety of objectives of the cited articles, i.e., short-term grid stability necessitates accuracy in magnitudes (MAE/RMSE) whereas day-ahead planning would benefit percentages to evaluate its errors (SMAPE/MAPE). Moreover, there is a considerable difference in the scale of the datasets used, about 24-h SCADA bursts and 17-year longitudinal data used by the ResTrans model. Such differences demonstrate that standard benchmark should be adopted in the next Solar PV studies so that they can directly compare models with each other.

Table 6 is exquisite in that it outlines the correlation between the transformer variant, optimal use, data type, forecast horizon, exceptional strength and appropriate uses case by case to enable the researcher to select the

Table 6 Suitability of transformer variants for different forecasting scenarios

References	Transformer Variant	Best Application	Type of Data Used	Forecast Horizon	Unique Strength	Suitable in case of
(Vaswani 2017)	Simple Transformer	Baseline PV forecasting	Time series	Short-term	Captures temporal dependencies	Small datasets, simple tasks
(Santos et al., 2022)	TFT	PV plant prediction	Time series, weather, and static	Day-ahead	Feature selection and interpretability	Heterogeneous inputs with multivariate weather variables
(Islam et al., 2023)	TFT	SCADA-based PV power	Time series and meteorological	Multi-step	Handles known future inputs	Multiple data sources are involved
(Zhan et al., 2024)	Patch-Transformer	Short-term PV forecasting	Patches of time series	15 min – 1 h	Local & global dependencies	High-resolution datasets
(Hu et al., 2024)	TSformer	Seasonal PV forecasting	Time series and weather	Multi-horizon	Models trend & seasonality	Highly seasonal patterns
(Wang & Kang, 2024)	WECT	Solar irradiance forecasting	NWP and weather	Short-term	Weather clustering	Weather-driven forecasts
(Huo et al., 2025)	CT-PatchTST	Long-term PV and wind	Multivariate channels	Hours to days	Inter-channel dependencies	Multi-sensor setups
(Mao et al., 2024)	GRU-Transformer	Short-term PV forecasting	Time series and meteorological	15 min – 1 h	Fast inference	Real-time forecasting needs
(Ziyabari et al., 2023)	ResTrans	Spatio-temporal forecasting	Satellite, SCADA, and weather	1–24 h	CNN and Transformer hybrid	Multiple PV sites
(Zhu et al., 2023b)	SL-Transformer	Noisy PV & wind prediction	Time series, LOF, and SG filters	24 h	Noise-resilient	Incomplete or noisy datasets

Table 7 Relative performance improvement of transformer variants over conventional models

Transformer variant	Benchmark task	Baseline model	Relative improvement (error reduction)	Resulting metric
Temporal Fusion Transformer (TFT)	Day-ahead PV power	LSTM/GRU	15% Reduction in MAE	$R^2 = 0.92$
Patch-Transformer	Short-term PV prediction	Standard Transformer	18% Reduction in MAE	Focused local learning
TSformer	Multi-horizon (1–24 h)	Non-decomposed DL models	22% Reduction in RMSE	Enhanced seasonal accuracy
CT-PatchTST	Long-term (1 week ahead)	Channel-independent (CI) models	20% Reduction in MAPE	Robust inter-channel modeling
GRU-Transformer	Intra-hour (15-min)	Standard GRU/RNN	12% Reduction in RMSE	High short-term precision
SL-Transformer	Day-ahead (Noisy data)	Standard Transformers	15% Gain in Efficiency	$R^2 = 0.9989$

right model of transformer to use in a given forecasting scenario (Liu et al., 2021).

Quantitative comparative analysis of performance gains

To compare the performance of Transformer variants, one needs to examine the improvements of these over the traditional forecasting models (e.g., LSTM, GRU, and Persistence models) in standardized predictive tasks, including day-ahead and intra-hour forecasting. Table 7 summarizes the error metrics reductions in the literature.

The results in Table 7 show that the large gains are obtained when models with domain specific structural priors are used. An example of this is TSformer 22% RMSE reduction, which can be mainly explained by the fact that it allows expressing the trend and seasonal data sources explicitly, which directly responds to the daily and seasonal oscillations of the solar data. Likewise, the 15% lowering of MAE by the TFT can be explained by the fact that the TFT uses a network with a variable selection, to select out meteorological noisy features that tend to worsen the performance of traditional LSTMs. These quantitative findings indicate that standard transformers support a robust temporal base, but the Second-Generation type (2023–2025) is required to deal with the non-stationarity of PV generation.

Future work

Development of transformer-based solar forecasting should not be confined to the general architectural articulations but need to focus on domain-specific innovations that would respond to the transparency-reliability-trade off observed in the current review. Future works are encouraged to look at the creation of Physics-Informed Attention Mechanisms (PIAM), in which the Clear Sky Models (CSM) are embedded as mathematics bias directly into the layer of self-attention. The proposed pathway makes the model follow the solar geometry $P_{Theoretical}$ to the extent that the solar geometry does not go beyond the physical limits in prediction like generation at night, which is still a loophole in the present model of black-boxes. Moreover, Conformal Prediction (CP) wrappers should be applied over grid operation architectures such as the Temporal Fusion Transformer (TFT), to enhance risk management in grid operations. This gives a strict mathematical basis to compute valid prediction intervals $\widehat{C}(x)$ with a user-specified coverage guarantee, which is effective in handling the non-stationary PV data without the computational load of the former Bayesian techniques.

Along the lines of the success of the Patch-Transformer, a radical methodological change must now be made towards Adaptive Multi-Scale Patching. Models can more effectively represent high-frequency flicker due to cloud transients by dynamically changing patch size Lp in time as the signal varies, than fixed-segment architectures (Chen et al., 2025). Although solutions such as TSformer and WECT have already been developed to incorporate Numerical Weather Prediction (NWP) data, the future of AI-driven weather forecasting efforts is in the area of Cross-Modal Fusion. Instead of direct concatenation of features, the next generation research ought to apply cross-attention blocks to map high-resolution weather tensors with the past generation tokens so that the model can be able to selectively attend to individual forecasted weather events to be predicted with more accuracy (Lu et al., 2024).

Finally, to solve the spatial challenges identified in multi-site studies like ResTrans, there is a clear need for Graph-Temporal Fusion Transformers. By treating individual PV farms as nodes within a graph and applying Graph Convolutional Networks (GCN) to extract spatial features before the transformer branch, researchers can create a concrete pathway for tracking cloud-front propagation across distributed energy networks (Ahmed et al., 2025). This integration of spatial graphs with temporal attention will facilitate the improvement of scalability and real-world deployment of PV forecasting systems in complex, multi-sensor environments.

Conclusion

Transformer-based architectures have been shown to outperform other designs in solar PV forecasting because they are sensitive to long-range temporal dependencies, multivariate input, and complex non-linear patterns not well represented by conventional models (Hertel et al., 2022). All of the reviewed variants Informer, TFT, Patch-Transformer, TSformer, and hybrid Transformer models, demonstrate a collection of improvements in interpretability, trend-seasonality decomposition, weather integration, and computational efficiency (Yang et al., 2024a) ((Sharma et al., 2022; Walczewski & Whrle, 2024)) (Tan et al., 2025).

Regardless of these developments, major issues persist, which include a lack of interpretability in certain models, a lack of uncertainty measures, and the inability to deal with non-stationary PV data (Ibrahim et al., 2024) (Salman et al., 2024). These gaps will be approached by the creation of adaptive transformer design, hybrid frameworks, and better methods of preprocessing data (Mao et al., 2024) (Massaoudi et al., 2021). The combination of numerical weather prediction (NWP) information and scalable architectures will enhance the accuracy and reliability of future forecasting systems even more (Sharma 2018; Al-Ali et al., 2023; Kumar et al., 2023).

Finally, the use of transformer-based models is a revolutionary approach to solar PV forecasting (Massaoudi et al., 2021). Work done in the future on the interpretability, and on the quality of the data, and on even better hybrid model work will enable full potential, so grids will be smarter and the future world where renewable energy will not be so vulnerable can be achieved (Massaoudi et al., 2021).

Competing interests

The authors declare that they have no competing interests.

Abbreviations

PV	Photovoltaic
AI	Artificial intelligence
NLP	Natural language processing
TFT	Temporal fusion transformer
RNN	Recurrent neural network
CNN	Convolutional neural network
LSTM	Long short-term memory
GRU	Gated recurrent unit
WECT	Weather embedding cluster transformer
TSformer	Decomposable transformer
NWP	Numerical weather prediction
ARIMA	Auto-regressive integrated moving average
LOF	Local outlier factor
SMAPE	Symmetric mean absolute percentage error
R2	Coefficient of determination
ResTrans	Multi-Branch ResNet-Transformer
SL-SL	Transformer
CT-PatchTST	Channel-Time Patch Time-Series Transformer
SG Filter	Savitzky–Golay Filter

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